1.3.6 ▶ ▶ Total Kinetic Energy of a Two-Body System

In this section, we use the result from 1.3.5 to derive an expression for the total energy of a two-body system. We specifically consider this system since it is the one under consideration for all two-body collisions.

Start by considering two masses m_1 and m_2 moving at velocities $\mathbf{v}_1$ and $\mathbf{v}_2$ respectively. Since we wish to find the sum $\sum \frac{1}{2} m_i v_{icm}^2$, we should find the velocity of each mass with respect to the velocity of their center of mass. By definition, the center of mass velocity is

$$\mathbf{v}_{cm} = \frac{m_1\mathbf{v}_1 + m_2\mathbf{v}_2}{m_1 + m_2}.$$

The velocity of m_1 with respect to the center of mass $\mathbf{v}_{1cm}$ is therefore

$$\begin{aligned}
\mathbf{v}_{1cm} &= \mathbf{v}_1 - \mathbf{v}_{cm} \\
&= \frac{m_1\mathbf{v}_1 + m_2\mathbf{v}_1 - m_1\mathbf{v}_1 - m_2\mathbf{v}_2}{m_1 + m_2} \\
&= \frac{m_2}{m_1 + m_2}(\mathbf{v}_1 - \mathbf{v}_2) \\
&= \frac{m_2}{m_1 + m_2}\mathbf{v}_{rel}
\end{aligned}$$

Where $\mathbf{v}_{rel} \equiv \mathbf{v}_1 - \mathbf{v}_2$ is the relative velocity of the two bodies. By symmetry, $\mathbf{v}_{2cm}$ equals

$$-\frac{m_1}{m_1 + m_2}\mathbf{v}_{rel}$$

Now, we can write the sum using these values

$$\begin{aligned}
\sum \frac{1}{2} m_i v_{icm}^2 &= \frac{1}{2} m_1 v_{1cm}^2 + \frac{1}{2} m_2 v_{2cm}^2 \\
&= \frac{1}{2}\frac{m_1 m_2^2}{(m_1 + m_2)^2} v_{rel}^2 + \frac{1}{2}\frac{m_1^2 m_2}{(m_1 + m_2)^2} v_{rel}^2 \\
&= \frac{1}{2}\frac{m_1 m_2}{m_1 + m_2} v_{rel}^2 \left(\frac{m_1 + m_2}{m_1 + m_2}\right) \\
&= \frac{1}{2}\mu v_{rel}^2
\end{aligned}$$

Where $\mu \equiv \frac{m_1 m_2}{m_1 + m_2}$ is the reduced mass of the system. Hence, the total kinetic energy of a two-body system is equal to

$$\boxed{K_{total} = \frac{1}{2} M v_{cm}^2 + \frac{1}{2}\mu v_{rel}^2} \tag{1.53}$$

This expression also provides an easy way to find the energy lost in a perfectly-inelastic collision, since v_{rel} after the collision is zero. Thus, the total energy lost in such a collision is $\frac{1}{2}\mu v_{rel}^2$ ★